

Use Case Descriptions by Subsystem – Doctor Appointment Management System

Subsystem: Account Management

Use case name : Register Account

Actors: Patient

Description: This use case allows a new patient to create an account in the system by providing required information like name, email, telephone, and password.

Preconditions: The patient is not already registered.

Postconditions: The patient account is successfully created and stored in the database.

Main Flow:

- 1.Patient opens registration page.
- 2.Fills out the required form fields.
- 3.Submits the form.
- 4.System validates the input and creates the account.
- 5.Confirmation message is shown.

Alternate Flows:

- If email is already registered, system shows an error.
- If required fields are missing, system prompts for completion.

Use case name: Login

Actors: Admin, Doctor, Patient

Description: This use case allows users (admin, doctor, patient) to log into the system using valid credentials to access their respective dashboards.

Preconditions: The user must be registered and have valid login credentials.

Postconditions: The user is authenticated and redirected to their role-specific dashboard.

Main Flow:

- 1.User navigates to the login page.
- 2.Enter email/username and password.
- 3.Submits the form.
- 4.System verifies the credentials.
- 5.User is redirected to their dashboard.

Alternate Flows:

- If credentials are invalid, system displays an error message.
- If fields are empty, system prompts the user to fill them.

Use case name: Update Profile

Actors: Admin, Doctor, Patient

Description: Allows users to update their personal information.

Preconditions: User must be logged in.

Postconditions: User data is updated in the system.

Subsystem: Appointment Booking

Use case name: View All Doctor

Actors: Patient

Description: This use case allows a patient to view the complete list of doctors available in the system. Each doctor profile includes name, specialization, contact information, and available working hours. This function helps the patient make an informed choice when selecting a doctor.

Preconditions: Patient must be logged into the system with a valid session.

Postconditions: The patient sees a comprehensive list of doctors and can take further action like scheduling.

Main Flow:

- 1.The patient logs in and navigates to the “Doctors” page.
- 2.The system queries the database for all registered doctors.
- 3.The system displays the list of doctors with their specialties, contact emails.
- 4.The patient can filter doctors by specialization or name.
- 5.The patient selects a doctor to view more detailed information or proceed to view available sessions.

Alternate Flows:

- If no doctors are found, the system displayed a “No doctors Available” message.
- If the system fails to retrieve data, an error message is shown with a retry option.

Business Rule:

- Only active doctors (approved by admin) are shown.

Use case name: View Scheduled Sessions

Actors: Admin, Patient, Doctor

Description: This use case enables a patient or doctor or admin to explore all upcoming scheduled sessions created by admin. Sessions include details like doctor name, doctor email, session title, session scheduled date, time. The patient can browse sessions to plan appointments.

Preconditions: Patient or doctor or admin must be logged into the system and there must be scheduled sessions added by admin.

Postconditions: The patient gets a clear view of available sessions and may proceed to book.

Main Flow:

- 1.Patient selects “Scheduled Sessions” from the dashboard.
- 2.The system fetches upcoming session data from the schedule entity.
- 3.All sessions are displayed, grouped by doctor or date.
- 4.Each session entry shows doctor name, time.
- 5.The patient may select filters (e.g., date range, doctor) to narrow the list.

Alternate Flow:

- If no sessions are scheduled, the system shows “No sessions available currently”.
- If the server fails, a retry prompt appears.

Use case name: Book a Session

Actor: Patient.

Description: This use case allows a patient to finalize a booking for a selected session from the list of scheduled sessions.

Preconditions: Patient must be logged in and the selected session must not be full or expired.

Postconditions: A booking is recorded in the system.

Main Flow:

1. The patient clicks on "Book Now" next to the desired session.
2. The system checks if the session is still available and has remaining capacity.
3. If available, the system registers the booking in the database and links it to the patient's profile.
4. The booking confirmation is displayed, along with session details and cancel.

Alternate Flows:

- If the session becomes full, the system prevents booking and shows a message.
- If patient tries to book a session that overlaps with an existing booking, the system alerts the patient.
- If the patient cancels midway, no changes are saved.

Business Rule:

- Patients cannot book multiple sessions at the same time.
- A doctor's session must be active to be bookable.

Use case name: Cancel Appointment

Actor: Patient, Admin

Description: Allows a patient and admin to cancel a previously booked appointment from their upcoming appointments list.

Preconditions:

- The patient must be logged in.
- The patient must have an active appointment scheduled in the future.

Postconditions:

- The Appointment is canceled successfully.
- The session becomes available again for booking.

Main Flow:

1. The patient navigates to "My Booking".
2. Selects booking from the upcoming list.
3. Clicks "Cancel Booking".
4. The system prompts for confirmation.

Alternate Flows:

-If the appointment is already past, the system disallows cancellation and displays a message.

-If the patient cancels too close to the session.

Subsystem: Schedule Management

Use case name: Add Schedule (Session)

Actors: Admin

Description: This use case allows the admin to define and add available working hours for doctors. These schedules become available for patients to view and book appointments.

Preconditions: Admin must be logged into the system.

Postconditions:

-A new schedule is created and associated with the doctor.

-It becomes visible to patients for booking.

Main Flow:

1. Admin navigates to the "Schedules" section.
2. Select "Add new Schedule (Session).
3. Input session title and chooses a doctor and input number of patient/Appointment numbers and inputs schedule date, start time.
4. Submits the form.
5. System validates and saves the schedule linked to the selected doctor.

Alternate Flows:

1. If the schedule overlaps with an existing one, system displays a conflict warning.
2. If fields are missing, system requests completion.

Use case name: Delete Schedule (Session)

Actors: Admin, Doctor

Description: Allows the Admin and doctor to delete an existing schedule that is no longer valid
Preconditions: Schedule exists and is not linked to any appointments.

Preconditions:

- The schedule must exist.

- It must not be linked to future appointments.

Postconditions: Schedule is successfully removed.

Main Flow:

1. Admin opens the "Schedule" section or doctor opens the "My sessions".
2. Selects the schedule to delete.
3. Confirms deletion request.
4. System deletes the schedule from the database.

Alternate Flows:

- If the schedule has active appointments, system prevents deletion and displays an error.